In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import DBSCAN

# Sample data
X = np.array([
    [1, 2], [1.5, 1.8], [2, 2],
    [8, 8], [8.5, 8], [9, 8.5],
    [20, 20]
])

# Create DBSCAN model
model = DBSCAN(eps=1.5, min_samples=2)

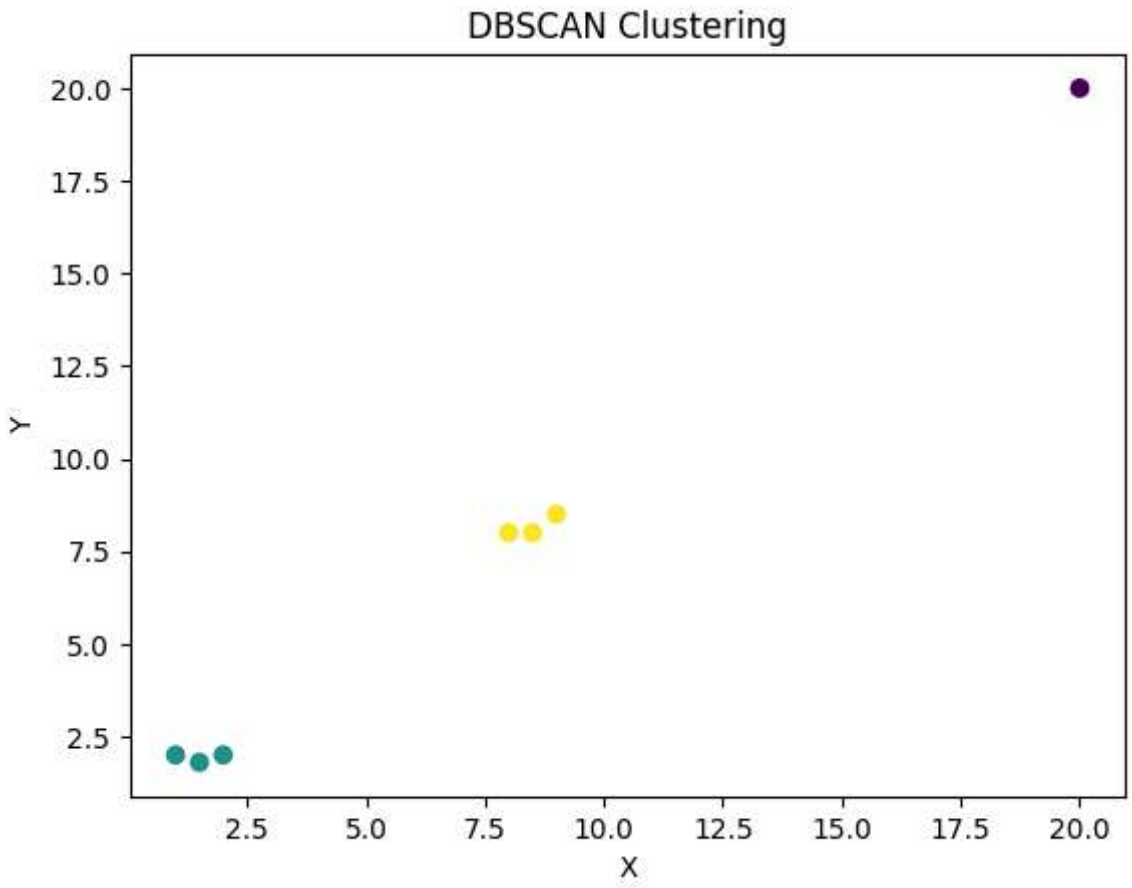
# Fit the model
labels = model.fit_predict(X)

print("Cluster labels:", labels)

# Plot clusters
plt.scatter(X[:, 0], X[:, 1], c=labels)

plt.xlabel("X")
plt.ylabel("Y")
plt.title("DBSCAN Clustering")
plt.show()
```

```
Cluster labels: [ 0  0  0  1  1  1 -1]
```


DBSCAN Clustering
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Y
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
X

In [ ]:

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